Teacher sheet lesson two

Introduction

Student information:

In this lesson you will be looking at the changes to the Australian continent. Our landscape has been formed through several geological occurrences. You will be working through two learning objects and collecting information from them in order to construct a timeline. This will need to be recorded in your science journal.

Task 1 — The timeline task

The timeline task is detailed in the document *Snapshots in time* which your teacher will give to you. You need to find photographs from the internet as learning object images cannot be copied. Work through the two activities on this page and gather evidence to include in your timeline. Use appropriate copyright and referencing practices.

Teacher information:

The activity *Snapshots in time* is found in the Queensland Studies Authority Assessment Bank. You will need to access this site and download a copy of this resource for students. In order to access the site you will need to register and receive a password. More information about this can be found on the teacher sheet *QSA Assessment Bank*.

The activity includes a student booklet and teacher guidelines. The student booklet needs to be provided to students. A link has been provided in the lesson overview and lesson plan for the QSA Assessment Bank homepage.

Task 2 — Learning object: Palaeotraveller

Student information:

Learning object one is called *Palaeotraveller* (https://learningplace.eq.edu.au/cx/resources/items/2b7ec0d6-14e4-bb27-3fc0-30c4953efb14/3/viewIMS.jsp). It provides information relevant to various locations at different points in time. It also contains an animated world map so you can see Australia's changing position during these time periods. Refer to the supplied instructions to guide you through this activity. Please do not spend more than 10 minutes on this activity. When working through this activity you need to think about:

a) what changes have occurred to the Australian continent?

b) what features of the present landscape give us clues?

Teacher information:

The learning object can be downloaded and saved into a common folder for the students to access. This may help with navigation speeds. A copy of the student instructions has been included with this document if you wish to use it as a handout.

Task 3 — Learning object: Travel back in time

Student information:

Use the learning object *Travel back in time* (https://learningplace.eq.edu.au/cx/resources/items/5a902fa5-4e1c-13f1-e92a-60c19ffbf3be/7/viewIMS.jsp?.hb=true) and the supplied instructions to make notes about evidence being used to support geological predictions. Select some of these pieces of evidence to incorporate into your timeline. Remember you may need to find other images from the internet. Please do not spend more then 10 minutes on this activity. In your science journal, write a sentence on how you think scientists have used evidence to support geological predications.

Teacher information:

The learning object can be downloaded and saved into a common folder for the students to access. This may help with navigation speeds. A copy of the student instructions has been included with this document if you wish to use it as a handout.

Instructions for *Palaeotraveller*

When you select the *Palaeotraveller* link, an introduction page will open up. You may click on the button to skip this. The home page will appear with a welcome message in the screen. Make sure the small window on the left of the screen reads ‘Cambrian’ (this is the earliest time period for this program) then press the go button.

The screen will now load with information about this period in time. On the green screen will appear a map of Australia. The red outline is the current coastline of the continent; click on the button that says key in order to understand the different areas of the map.

Open up a blank Word document and create a table using the following template. You will have more room if you change your page setup to landscape.

Time period	Position of continent on planet	Temp	Sea level	What did the continent of Australia look like during this period?	What are the main changes to the coastline from the previous period?	Where is the green cross? What does the landscape look like here?	Description of life forms

Now work your way through several of the time periods, filling in the table with your information as you go. You will use this information to construct your timeline. Save your document to your personal folder or record in your science journal.

Instructions for *Travel back in time*

Open the learning object and press the continue button at the bottom of the screen until you are asked to choose a collection. You will be given the opportunity to work through all four collections so it does not matter which one you choose first.

Once you select a collection, four fossils will appear on the floating platforms. It will be your job to label them by researching the information on the website. Press the continue button.

A time portal will open up on the screen. You can choose any of the fossils and drag them into the blue section of the time portal (the label will follow the fossil). You will then be shown a screen that has a Time Portal console at the top, then some instructions written in yellow and finally the fossil and label of the bottom of the screen.

If you wish to enlarge the label, click on the cross on the top right. You can keep working with it enlarged.

You will be given one piece of information on the label. You will need to use this to fill in the rest of the information. Each drop-down box will give you several choices.

When you have found the correct organism, make the correct selections in the drop-down boxes. Then click on the picture of the organism and a picture should appear in the space on the right-hand side of the label.

Your can then copy and paste the information on the organism into the white window on the label. It will expand to fit the information.

Your label is finished now; press the button that says exit the time portal. This will take you back to the previous screen so you can select another fossil.